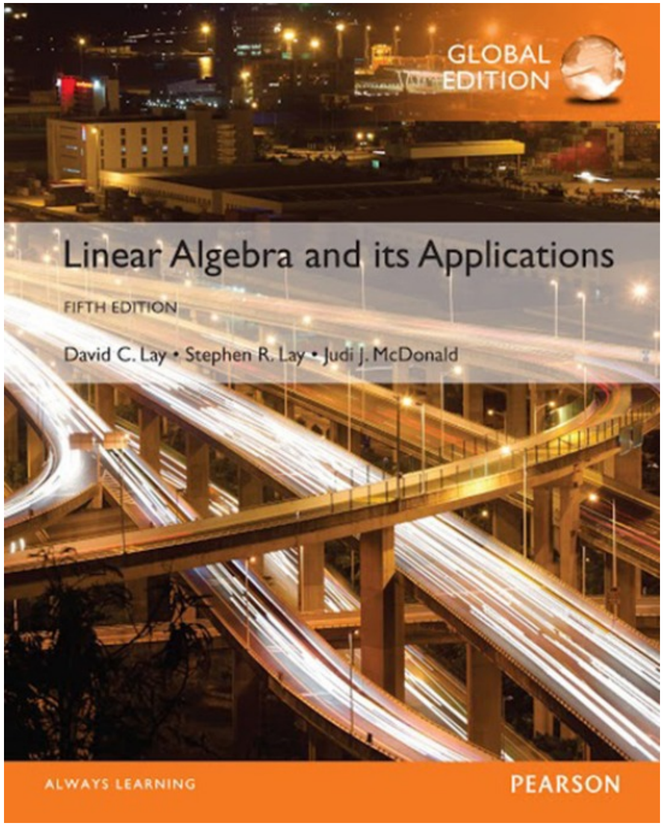
7 Symmetric Matrices and Quadratic Forms

7.1

DIAGONALIZATION OF SYMMETRIC MATRICES



PEARSON

SYMMETRIC MATRIX

- A **symmetric matrix** is a matrix A such that $A^T = A$.
- Such a matrix is necessarily square.
- Its main diagonal entries are arbitrary, but its other entries occur in pairs—on opposite sides of the main diagonal.

SYMMETRIC MATRIX

- **Theorem 1:** If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.
- **Proof:** Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors that correspond to distinct eigenvalues, say, λ_1 and λ_2.
- To show that $v_1.v_2 = 0$, compute

$$\lambda_1 v_1.v_2 = (\lambda_1 v_1)^T v_2 = (Av_1)^T v_2$$ Since $\mathbf{v}_1$ is an eigenvector

$$= (v_1^T A^T) v_2 = v_1^T (Av_2)$$ Since $A^T = A$

$$= v_1^T (\lambda_2 v_2)$$ Since $\mathbf{v}_2$ is an eigenvector

$$= \lambda_2 v_1^T v_2 = \lambda_2 v_1.v_2$$

SYMMETRIC MATRIX

- Hence $(\lambda_1 - \lambda_2)\, v_1 . v_2 = 0$
- But $\lambda_1 - \lambda_2 \neq 0$, so $v_1 . v_2 = 0$.
- An $n \times n$ matrix A is said to be **orthogonally diagonalizable** if there are an orthogonal matrix P (with $P^{-1} = P^T$) and a diagonal matrix D such that

(1) $$A = PDP^T = PDP^{-1}$$

- Such a diagonalization requires n linearly independent and orthonormal eigenvectors.
- When is this possible?
- If A is orthogonally diagonalizable as in (1), then

$$A^T = (PDP^T)^T = P^{TT} D^T P^T = PDP^T = A$$

SYMMETRIC MATRIX

- Thus A is symmetric!
- **Theorem 2**: An $n \times n$ matrix A is orthogonally diagonalizable if and only if A is symmetric matrix.

- **Example 3:** Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$, whose characteristic equation is

$$0 = -\lambda^3 + 12\lambda^2 - 21\lambda - 98 = -(\lambda - 7)^2(\lambda + 2)$$

SYMMETRIC MATRIX

- **Solution:** The usual calculations produce bases for the eigenspaces:

$$\lambda = 7 : \mathrm{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathrm{v}_2 = \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix}; \lambda = -2 : \mathrm{v}_3 = \begin{bmatrix} -1 \\ -1/2 \\ 1 \end{bmatrix}$$

- Although $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent, they are not orthogonal.
- The projection of $\mathbf{v}_2$ onto $\mathbf{v}_1$ is $\frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1$.

SYMMETRIC MATRIX

- The component of $\mathbf{v}_2$ orthogonal to $\mathbf{v}_1$ is

$$z_2 = v_2 - \frac{v_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} - \frac{-1/2}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/4 \\ 1 \\ 1/4 \end{bmatrix}$$

- Then $\{\mathbf{v}_1, \mathbf{z}_2\}$ is an orthogonal set in the eigenspace for $\lambda = 7$.

- (Note that $\mathbf{z}_2$ is linear combination of the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$, so $\mathbf{z}_2$ is in the eigenspace).

SYMMETRIC MATRIX

- Since the eigenspace is two-dimensional (with basis $\mathbf{v}_1$, $\mathbf{v}_2$), the orthogonal set $\{\mathbf{v}_1, \mathbf{z}_2\}$ is an *orthogonal basis* for the eigenspace, by the Basis Theorem.

- Normalize $\mathbf{v}_1$ and $\mathbf{z}_2$ to obtain the following orthonormal basis for the eigenspace for $\lambda = 7$:

$$u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}, u_2 = \begin{bmatrix} -1/\sqrt{18} \\ 4/\sqrt{18} \\ 1/\sqrt{18} \end{bmatrix}$$

SYMMETRIC MATRIX

- An orthonormal basis for the eigenspace for $\lambda = -2$ is

$$u_3 = \frac{1}{\|2v_3\|} 2v_3 = \frac{1}{3}\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2/3 \\ -1/3 \\ 2/3 \end{bmatrix}$$

- By Theorem 1, $\mathbf{u}_3$ is orthogonal to the other eigenvectors $\mathbf{u}_1$ and $\mathbf{u}_2$.

- Hence $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal set.

SYMMETRIC MATRIX

- Let

$$P = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{18} & -2/3 \\ 0 & 4/\sqrt{18} & -1/3 \\ 1/\sqrt{2} & 1/\sqrt{18} & 2/3 \end{bmatrix}, D = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

- Then P orthogonally diagonalizes A, and $A = PDP^{-1}$.

THE SPECTRAL THEOREM

- The set if eigenvalues of a matrix A is sometimes called the *spectrum* of A, and the following description of the eigenvalues is called a *spectral theorem*.

- **Theorem 3:** The Spectral Theorem for Symmetric Matrices
- An $n \times n$ symmetric matrix A has the following properties:
  a. A has n real eigenvalues, counting multiplicities.

THE SPECTRAL THEOREM

b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ as a root of the characteristic equation.

c. The eigenspaces are mutually orthogonal, in the sense that eigenvectors corresponding to different eigenvalues are orthogonal.

d. A is orthogonally diagonalizable.

SPECTRAL DECOMPOSITION

- Suppose $A = PDP^{-1}$, where the columns of P are orthonormal eigenvectors $\mathbf{u}_1,\ldots,\mathbf{u}_n$ of A and the corresponding eigenvalues $\lambda_1,\ldots,\lambda_n$ are in the diagonal matrix D.
- Then, since $P^{-1} = P^T$,

$$A = PDP^T = \begin{bmatrix} \mathrm{u}_1 & \cdots & \mathrm{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathrm{u}_1^T \\ \vdots \\ \mathrm{u}_n^T \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 \mathrm{u}_1 & \cdots & \lambda_n \mathrm{u}_n \end{bmatrix} \begin{bmatrix} \mathrm{u}_1^T \\ \vdots \\ \mathrm{u}_n^T \end{bmatrix}$$

SPECTRAL DECOMPOSITION

- Using the column-row expansion of a product, we can write

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \cdots + \lambda_n \mathbf{u}_n \mathbf{u}_n^T \qquad (2)$$

- This representation of A is called a **spectral decomposition** of A because it breaks up A into pieces determined by the spectrum (eigenvalues) of A.
- Each term in (2) is an $n \times n$ matrix of rank 1.
- For example, every column of $\lambda_1 \mathbf{u}_1 \mathbf{u}_1^T$ is a multiple of $\mathbf{u}_1$.
- Each matrix $\mathbf{u}_j \mathbf{u}_j^T$ is a **projection matrix** in the sense that for each $\mathbf{x}$ in $\mathbb{R}^n$, the vector $(\mathbf{u}_j \mathbf{u}_j^T)\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto the subspace spanned by $\mathbf{u}_j$.

SPECTRAL DECOMPOSITION

- **Example 4:** Construct a spectral decomposition of the matrix A that has the orthogonal diagonalization

$$A = \begin{bmatrix} 7 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

- **Solution:** Denote the columns of P by $\mathbf{u}_1$ and $\mathbf{u}_2$.

- Then $$A = 8\mathrm{u}_1\mathrm{u}_1^T + 3\mathrm{u}_2\mathrm{u}_2^T$$

SPECTRAL DECOMPOSITION

- To verify the decomposition of A, compute

$$\mathbf{u}_1\mathbf{u}_1^T = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{bmatrix}$$

$$\mathbf{u}_2\mathbf{u}_2^T = \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{bmatrix}$$

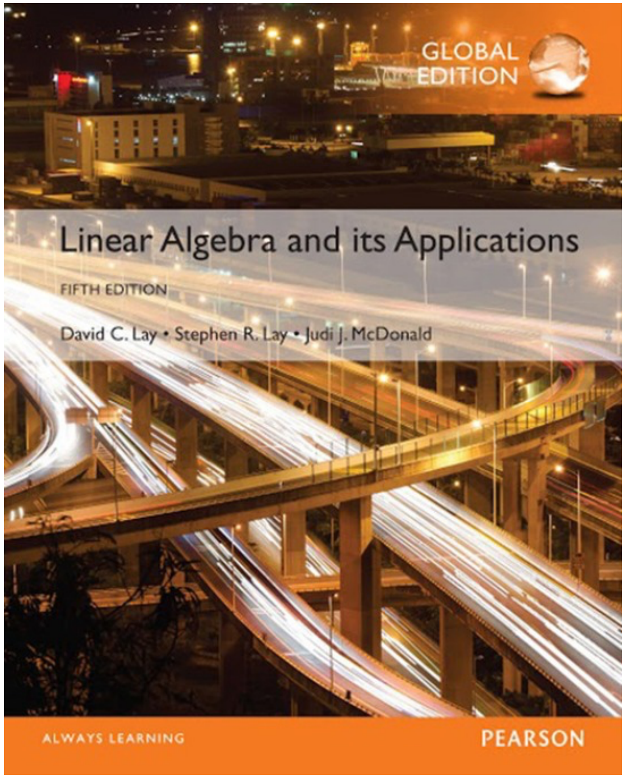
and

$$8\mathbf{u}_1\mathbf{u}_1^T + 3\mathbf{u}_2\mathbf{u}_2^T = \begin{bmatrix} 32/5 & 16/5 \\ 16/5 & 8/5 \end{bmatrix} + \begin{bmatrix} 3/5 & -6/5 \\ -6/5 & 12/5 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 2 & 4 \end{bmatrix} = A$$

7 Symmetric Matrices and Quadratic Forms

7.2

QUADRATIC FORMS



QUADRATIC FORMS

- A **quadratic form** on $\mathbb{R}^n$ is a function Q defined on $\mathbb{R}^n$ whose value at a vector **x** in $\mathbb{R}^n$ can be computed by an expression of the form $Q(\mathrm{x}) = \mathrm{x}^T A\mathrm{x}$, where A is an $n \times n$ symmetric matrix.

- The matrix A is called the **matrix of the quadratic form**.

QUADRATIC FORMS

- **Example 1:** Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Compute $\mathbf{x}^T A \mathbf{x}$ for the following matrices.

  a. $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$

  b. $A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$

QUADRATIC FORMS

- **Solution:**

a. $\mathrm{x}^T A\mathrm{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4x_1 \\ 3x_2 \end{bmatrix} = 4x_1^2 + 3x_2^2.$

b. There are two -2 entries in A.

$$\mathrm{x}^T A\mathrm{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3x_1 - 2x_2 \\ -2x_1 + 7x_2 \end{bmatrix}$$

$$= x_1(3x_1 - 2x_2) + x_2(-2x_1 + 7x_2)$$

$$= 3x_1^2 - 2x_1x_2 - 2x_2x_1 + 7x_2^2$$

$$= 3x_1^2 - 4x_1x_2 + 7x_2^2$$

QUADRATIC FORMS

- The presence of $-4x_1x_2$ in the quadratic form in Example 1(b) is due to the -2 entries off the diagonal in the matrix A.

- In contrast, the quadratic form associated with the diagonal matrix A in Example 1(a) has no x_1x_2 *cross-product* term.

CHANGE OF VARIBALE IN A QUADRATIC FORM

- If **x** represents a variable vector in $\mathbb{R}^n$, then a **change of variable** is an equation of the form

$$\mathrm{x} = P\mathrm{y}, \text{ or equivalently, } \mathrm{y} = P^{-1}\mathrm{x} \qquad (1)$$

  where P is an invertible matrix and **y** is a new variable vector in $\mathbb{R}^n$.

- Here **y** is the coordinate vector of **x** relative to the basis of $\mathbb{R}^n$ determined by the columns of P.

- If the change of variable (1) is made in a quadratic form $\mathbf{x}^T A \mathbf{x}$, then

$$(2) \quad \mathrm{x}^T A\mathrm{x} = (P\mathrm{y})^T A(P\mathrm{y}) = \mathrm{y}^T P^T AP\mathrm{y} = \mathrm{y}^T (P^T AP)\mathrm{y}$$

  and the new matrix of the quadratic form is P^TAP.

CHANGE OF VARIBALE IN A QUADRATIC FORM

- Since A is symmetric, Theorem 2 guarantees that there is an *orthogonal* matrix P such that P^TAP is a diagonal matrix D, and the quadratic form in (2) becomes $\mathbf{y}^TD\mathbf{y}$.

- **Example 4:** Make a change of variable that transforms the quadratic form $Q(\mathrm{x}) = x_1^2 - 8x_1x_2 - 5x_2^2$ into a quadratic form with no cross-product term.

- **Solution:** The matrix of the given quadratic form is

$$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}$$

CHANGE OF VARIBALE IN A QUADRATIC FORM

- The first step is to orthogonally diagonalize A.
- Its eigenvalues turn out to be $\lambda = 3$ and $\lambda = -7$.
- Associated unit eigenvectors are

$$\lambda = 3 : \begin{bmatrix} 2/\sqrt{5} \\ -1/\sqrt{5} \end{bmatrix}; \lambda = -7 : \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

- These vectors are automatically orthogonal (because they correspond to distinct eigenvalues) and so provide an orthonormal basis for $\mathbb{R}^2$.

CHANGE OF VARIBALE IN A QUADRATIC FORM

- Let

$$P=\begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}, D=\begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix}$$

- Then $A=PDP^{-1}$ and $D=P^{-1}AP=P^{T}AP.$

- A suitable change of variable is

$$\mathrm{x}=P\mathrm{y}\text{, where } \mathrm{x}=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ and } \mathrm{y}=\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

CHANGE OF VARIBALE IN A QUADRATIC FORM

- Then

$$\begin{aligned} x_1^2 - 8x_1x_2 - 5x_2^2 &= \mathrm{x}^T A\mathrm{x} = (P\mathrm{y})^T A(P\mathrm{y}) \\ &= \mathrm{y}^T P^T AP\mathrm{y} = \mathrm{y}^T D\mathrm{y} \\ &= 3\mathrm{y}_1^2 - 7\mathrm{y}_2^2 \end{aligned}$$

- To illustrate the meaning of the equality of quadratic forms in Example 4, we can compute $Q(\mathbf{x})$ for $\mathrm{x} = (2, -2)$ using the new quadratic form.

CHANGE OF VARIBALE IN A QUADRATIC FORM

- First, since $\mathbf{x} = P\mathbf{y}$,

$$\mathbf{y} = P^{-1}\mathbf{x} = P^{T}\mathbf{x}$$

so

$$y = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 6/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$$

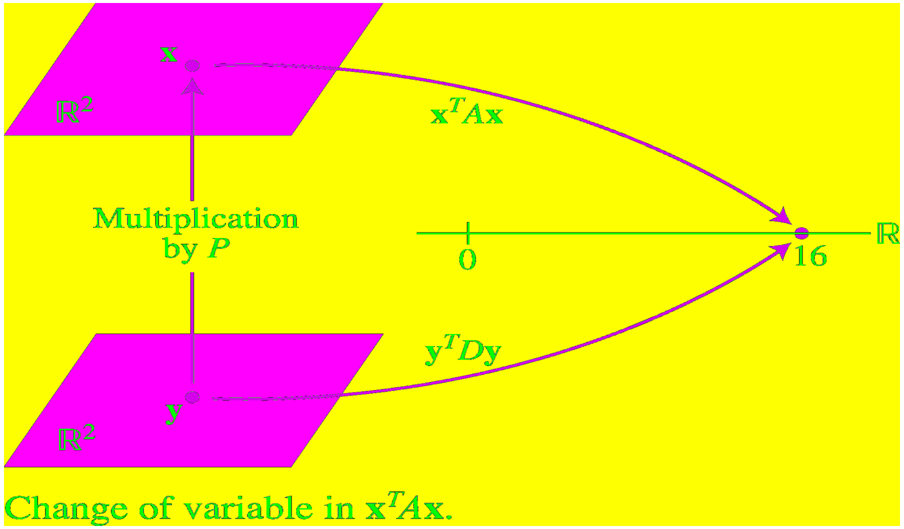
- Hence

$$3y_1^2 - 7y_2^2 = 3(6/\sqrt{5})^2 - 7(-2/\sqrt{5})^2 = 3(36/5) - 7(4/5)$$
$$= 80/5 = 16$$

- This is the value of $Q(\mathbf{x})$ when $\mathbf{x} = (2, -2)$.

THE PRINCIPAL AXIS THEOREM

- See the figure below.



- **Theorem 4**: Let A be an $n \times n$ symmetric matrix. Then there is an orthogonal change of variable, $\mathrm{x} = P\mathrm{y}$, that transforms the quadratic form $\mathbf{x}^T A\mathbf{x}$ into a quadratic form $\mathbf{y}^T D\mathbf{y}$ with no cross-product term.

THE PRINCIPAL AXIS THEOREM

- The columns of P in theorem 4 are called the **principal axes** of the quadratic form $\mathbf{x}^T A\mathbf{x}$.

- The vector $\mathbf{y}$ is the coordinate vector of $\mathbf{x}$ relative to the orthonormal basis of $\mathbb{R}^n$ given by these principal axes.

A Geometric View of Principal Axes

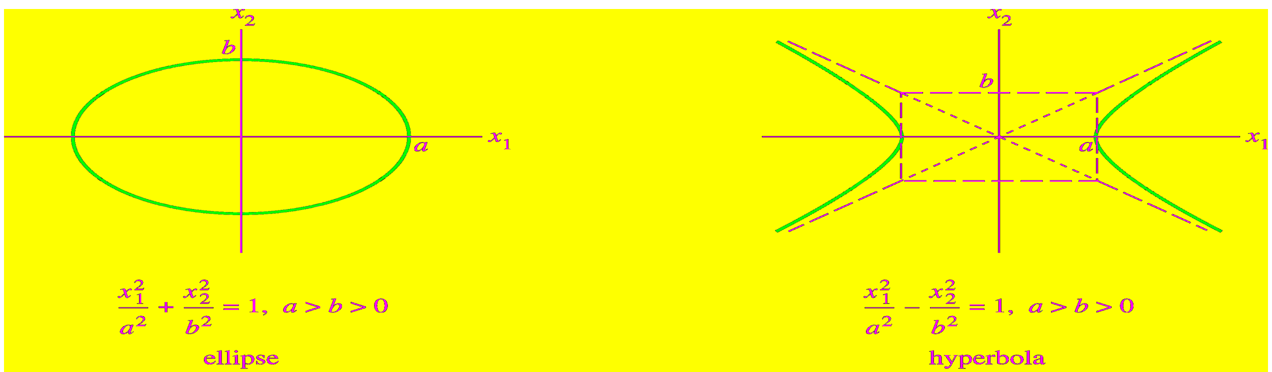
- Suppose $Q(\mathrm{x}) = \mathrm{x}^T A\mathrm{x}$, where A is an invertible 2×2 symmetric matrix, and let c be a constant.

A GEOMETRIC VIEW OF PRINCIPAL AXES

- It can be shown that the set of all $\mathbf{x}$ in $\mathbb{R}^2$ that satisfy

$$\mathrm{x}^T A\mathrm{x} = c \qquad (3)$$

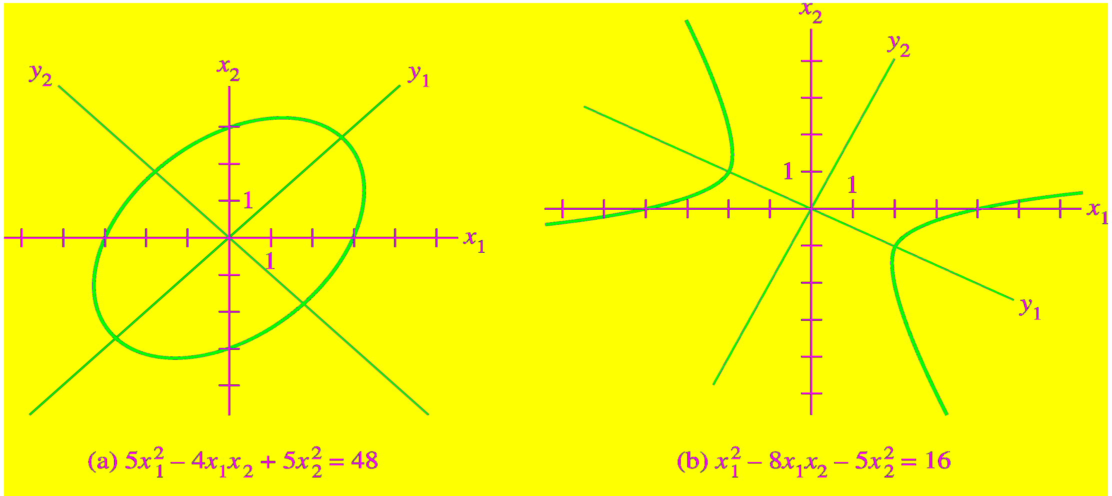
  either corresponds to an ellipse (or circle), a hyperbola, two intersecting lines, or a single point, or contains no points at all.
- If A is a diagonal matrix, the graph is in *standard position*, such as in the figure below.



An ellipse and a hyperbola in standard position.

A GEOMETRIC VIEW OF PRINCIPAL AXES

- If A is not a diagonal matrix, the graph of equation (3) is rotated out of standard position, as in the figure below.



An ellipse and a hyperbola *not* in standard position.

- Finding the *principal axes* (determined by the eigenvectors of A) amounts to finding a new coordinate system with respect to which the graph is in standard position.

CLASSIFYING QUADRATIC FORMS

- **Definition:** A quadratic form Q is:
  - **a. positive definite** if $Q(\mathrm{x}) > 0$ for all $\mathrm{x} \neq 0$,
  - **b. negative definite** if $Q(\mathrm{x}) < 0$ for all $\mathrm{x} \neq 0$,
  - **c. indefinite** if $Q(\mathbf{x})$ assumes both positive and negative values.

- Also, Q is said to be **positive semidefinite** if $Q(\mathrm{x}) \geq 0$ for all **x**, and **negative semidefinite** if $Q(\mathrm{x}) \leq 0$ for all **x**.

QUADRATIC FORMS AND EIGENVALUES

- **Theorem 5:** Let A be an $n \times n$ symmetric matrix. Then a quadratic form $\mathbf{x}^T A \mathbf{x}$ is:
  a. positive definite if and only if the eigenvalues of A are all positive,
  b. negative definite if and only if the eigenvalues of A are all negative, or
  c. indefinite if and only if A has both positive and negative eigenvalues.

QUADRATIC FORMS AND EIGENVALUES

- **Proof:** By the Principal Axes Theorem, there exists an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$ such that

$$Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} = \mathbf{y}^T D\mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \quad (4)$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A.

- Since P is invertible, there is a one-to-one correspondence between all nonzero **x** and all nonzero **y**.

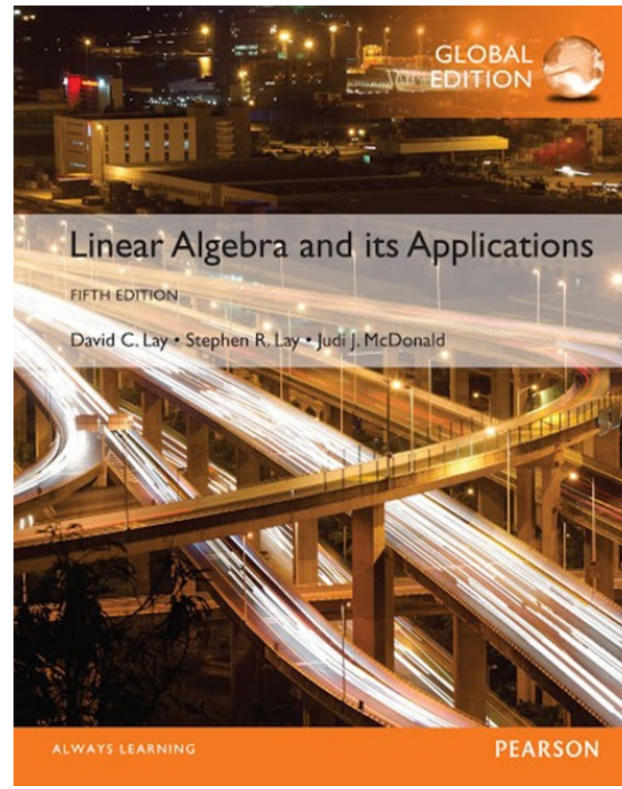
QUADRATIC FORMS AND EIGENVALUES

- Thus the values of $Q(\mathbf{x})$ for $\mathbf{x} \neq 0$ coincide with the values of the expression on the right side of (4), which is controlled by the signs of the eigenvalues $\lambda_1, \ldots, \lambda_n$, in three ways described in the theorem.

7 Symmetric Matrices and Quadratic Forms

7.3

CONSTRAINED OPTIMIZATION



CONSTRAINED OPTIMIZATION

- **Theorem 6**Let A be a symmetric matrix, and define m and M as in (2). Then M is the greatest eigenvalue λ_1 of A and m is the least eigenvalue of A. The value of $x^T A x$ is M when x is a unit eigenvector u_1 corresponding to M. The value of $x^T A x$ is m when x is a unit eigenvector corresponding to m.

- **Proof** Orthogonally diagonalize A as PDP^{-1}. We know that

$$x^T A x = yTDy \quad \text{when} \quad x = Py \qquad (3)$$

CONSTRAINED OPTIMIZATION

- Also,

$$\|x\| = \|Py\| = \|y\| \text{ for all y}$$

- Because $P^TP = I$ and $\|Py\|^2 = (Py)^T(Py) = yTP^TPy = yTy = \|y\|^2$. In particular, $\|y\| = 1$ if and only if $\|x\| = 1$. Thus, x^TAx and y^TDyassume the same set of values as x and y range over the set of all unit vectors.

- To simplify notation, suppose that A is a 3×3 matrix with eigenvalues $a \leq b \leq c$. Arrange the columns of P so that $P = [u_1 \;\; u_2 \;\; u_3]$ and

CONSTRAINED OPTIMIZATION

$$D = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

- Given any unit vector y in $\mathbb{R}^3$ with coordinates y_1, y_2, y_3, observe that

$$\begin{aligned} ay_1^2 &= ay_1^2 \\ by_2^2 &\le ay_2^2 \\ cy_3^2 &\le ay_3^2 \end{aligned}$$

- and obtain these inequalities:

$$\begin{aligned} y^T D y &= ay_1^2 + by_2^2 + cy_3^2 \\ &\le ay_1^2 + ay_2^2 + ay_3^2 \\ &= a(y_1^2 + y_2^2 + y_3^2) = a\|y\|2 = a \end{aligned}$$

CONSTRAINED OPTIMIZATION

- Thus $M \leq a$, by definition of M. However, $y^T Dy = a$ when $y = e_1 = (1, 0, 0)$, so in fact $M = a$. By (3), the x that corresponds by $y = e_1$ is the eigenvector u_1 of A, because

$$x = Pe_1 = [u_1 \quad u_2 \quad u_3]\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = u_1$$

- Thus $M = a = e_1^T De_1 = u_1^T Au_1$, which proves the statement about M. A similar argument shows that m is the least eigenvalue, c, and this value of $x^T Ax$ is attained when $x = Pe_3 = u_3$.

CONSTRAINED OPTIMIZATION

- **Example 3** Let $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$. Find the maximum value of the quadratic form $\mathbf{x}^T A\mathbf{x}$ subject to the constraint $\mathbf{x}^T\mathbf{x} = 1$, and find a unit vector at which this maximum value is attained.
- **Solution** By Theorem 6, the desired maximum value is the greatest eigenvalue of A. The characteristic equation turns out to be

$$0 = -\lambda^3 + 10\lambda^2 - 27\lambda + 18 = -(\lambda - 6)(\lambda - 3)(\lambda - 1)$$

CONSTRAINED OPTIMIZATION

- The greatest eigenvalue is 6.
- The constrained maximum of $x^T A x$ is attained when x is a unit eigenvector for $\lambda = 6$. Solve $(A - 6I)x = 0$ and find an eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Set $u_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$.

CONSTRAINED OPTIMIZATION

- **Theorem 7** Let A, λ_1, and u_1 be as in Theorem 6. Then the maximum value of x^TAx subject to the constraints

$$x^Tx = 1, x^Tu_1 = 0$$

- is the second greatest eigenvalue λ_2, and this maximum is attained when x is an eigenvector u_2 corresponding to λ_2.

- **Example 4** Find the maximum value of $9x_1^2 + 4x_2^2 + 3x_3^2$ subject to the constraints $x^Tx = 1$,and $x^Tu_1 = 0$, where $u_1 = (1, 0, 0)$. Note that u_1 is a unit eigenvector corresponding to the greatest eigenvalue $\lambda = 9$ of the matrix of the quadratic form.

CONSTRAINED OPTIMIZATION

- **Solution** If the coordinates of x are x_1, x_2, x_3, then the constraint $x^T u_1 = 0$ means simply that $x_1 = 0$. For such a unit vector, $x_2^2 + x_3^2 = 1$, and

$$
\begin{aligned}
9x_1^2 + 4x_2^2 + 3x_3^2 &= 4x_2^2 + 3x_3^2 \\
&\leq 4x_2^2 + 4x_3^2 \\
&= 4(x_2^2 + x_3^2) \\
&= 4
\end{aligned}
$$

- Thus the constrained maximum of the quadratic form does not exceed 4. And this value is attained for x = (0, 1, 0) which is the eigenvector for the second greatest eigenvalue of the matrix of the quadratic form.

CONSTRAINED OPTIMIZATION

- **Theorem 8** Let A be a symmetric $n \times n$ matrix with an orthogonal diagonalization $A = PDP^{-1}$, where the entries on the diagonal of D are arranged so that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ and where columns of P are corresponding unit eigenvectors $u_1, \ldots, u_n$. Then for $k = 2, \ldots, n$, the maximum value of $x^T A x$ subject to the constraints

$$x^T x = 1, x^T u_1 = 0, \quad \ldots, \quad x^T u_{k-1} = 0$$

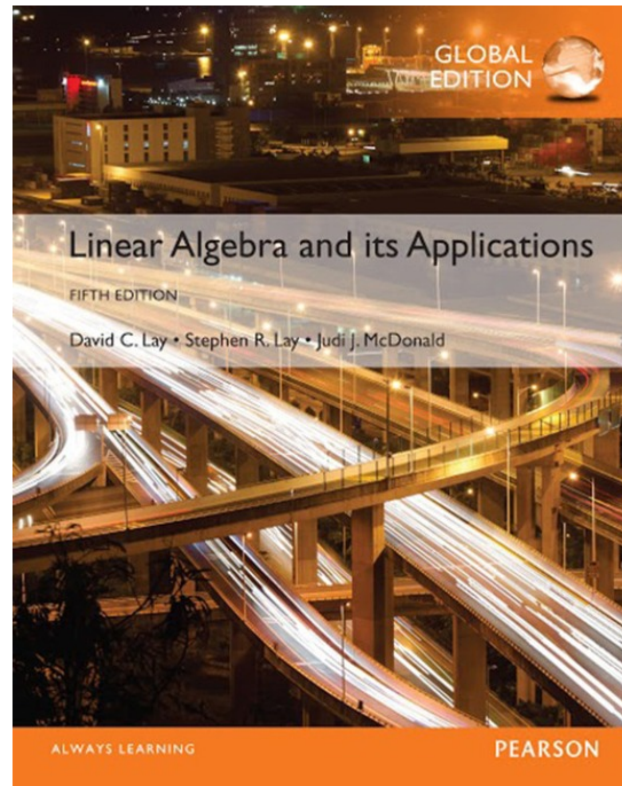
- is the eigenvalue λ_k, and this maximum is attained at $x = u_k$.

7

Symmetric Matrices and Quadratic Forms

7.4

THE SINGULAR VALUE DECOMPOSITION



THE SINGULAR VALUES OF AN $m \times n$ MATRIX

- **Theorem 9** Suppose $\{v_1, \ldots, v_n\}$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A^TA, arranged so that the corresponding eigenvalues of A^TA satisfy $\lambda_1 \geq \cdots \geq \lambda_n$, and suppose A has r nonzero singular values. Then $\{Av_1, \ldots, Av_r\}$ is an orthogonal basis for Col A, and rank $A = r$.

- **Proof** Because v_i and $\lambda_j v_j$ are orthogonal for $i \neq j$,
$$(Av_i)^T(Av_j) = v_i^T A^T A v_j = v_i^T(\lambda_j v_j) = 0$$

- Thus $\{Av_1, \ldots, Av_n\}$ is an orthogonal set.

THE SINGULAR VALUES OF AN $m \times n$ MATRIX

- Since the lengths of the vectors $Av_1, \ldots, Av_n$ are the singular values of A, and since there are r nonzero singular values, $Av_i \neq 0$ if and only if $1 \leq i \leq r$.
- So $Av_1, \ldots, Av_r$ are linearly independent vectors, and they are in Col A.
- Finally, for any y in Col A—say, y = Ax—we can write $x = c_1v_1 + \cdots + cnvn$, and

$$y = Ax = c_1Av_1 + \cdots + crAv_r + cr_{+1}Av_{r+1} + \cdots + cnAv_n$$

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$$= c_1 A v_1 + \cdots + cr A v_r + 0 + \cdots + 0$$

- Thus **y** is in Span$\{Av_1, \ldots, A\mathrm{v}_r\}$, which shows that $\{Av_1, \ldots, A\mathrm{v}_r\}$ is an (orthogonal) basis for Col A.

- Hence rank A = dim Col A = r.

THE SINGULAR VALUE DECOMPOSITION

- **Theorem 10: The Singular Value Decomposition** Let A be an $m \times n$ matrix with rank r. Then there exists an $m \times n$matrix Σ as in (3) for which the diagonal entries in D are the first r singular values of A, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, and there exist an $m \times m$ orthogonal matrix U and an $n \times n$ orthogonal matrix V such that

$$A = U\Sigma V^T$$

THE SINGULAR VALUE DECOMPOSITION

- Any factorization $A = U\Sigma V^T$, with U and V orthogonal, Σ as in (3), and positive diagonal entries in D, is called a **singular value decomposition** (or **SVD**) of A.

- The columns of U in such a decomposition are called **left singular vectors** of A, and the columns of V are called **right singular vectors** of A.

- **Proof** Let λ_i and v_i be as in Theorem 9, so that $\{Av_1, \ldots, A\mathrm{v}_\mathrm{r}\}$ is an orthogonal basis for Col A.

THE SINGULAR VALUE DECOMPOSITION

- Normalize each $A\mathrm{v}_i$ to obtain an orthonormal basis $\{\mathrm{u}_1, \ldots, \mathrm{u}_r\}$, where

$$u_i = \frac{1}{\|Av_i\|} Av_i = \frac{1}{\sigma_1} Av_i$$

- And

$$Av_i = \sigma_i u_i \qquad (1 \le i \le r) \qquad (4)$$

- Now extend $\{\mathrm{u}_1, \ldots, \mathrm{u}_r\}$ to an orthonormal basis $\{\mathrm{u}_1, \ldots, \mathrm{u}_m\}$ of $\mathbb{R}^m$, and let

$$U = [u_1 \;\; u_2 \;\ldots\; um] \quad \text{and } V = [v_1 \;\; v_2 \;\ldots\; vm]$$

- By construction, U and V are orthogonal matrices.

THE SINGULAR VALUE DECOMPOSITION

- Also, from (4),

  $AV = [Ax_1 \;\; \ldots \;\; Avr \;\;\; 0 \;\; \ldots \;\; 0] = [\sigma_1 u_1 \ldots \sigma_r u_r \;\; 0 \ldots 0]$

- Let D be the diagonal matrix with diagonal entries $\sigma_1, \ldots, \sigma_r$, and let Σ be as in (3) above. Then

$$U\Sigma = [\,\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_m\,] \left[\begin{array}{cccc|c} \sigma_1 & & & 0 & \\ & \sigma_2 & & & 0 \\ & & \ddots & & \\ 0 & & & \sigma_r & \\ \hline & & 0 & & 0 \end{array}\right]$$

$$= [\,\sigma_1\mathbf{u}_1 \quad \cdots \quad \sigma_r\mathbf{u}_r \quad 0 \quad \cdots \quad 0\,]$$
$$= AV$$

- Since V is an orthogonal matrix, $U\Sigma V^T = AVVT = A$.

THE SINGULAR VALUE DECOMPOSITION

- **Example 3** Use the results of Examples 1 and 2 to construct a singular value decomposition of $A = \begin{bmatrix} 4 & 11 & 14 \\ 8 & 7 & -2 \end{bmatrix}$.

- **Solution** A construction can be divided into three steps.
- ***Step 1. Find an orthogonal diagonalization of A^TA***. That is, find the eigenvalues of A^TA and a corresponding orthonormal set of eigenvectors. If A had only two columns, the calculations could be done by hand. Larger matrices usually require a matrix program. However, for the matrix A here, the eigendata for A^TA are provided in Example 2.

THE SINGULAR VALUE DECOMPOSITION

- ***Step 2*. *Set up V and Σ*.** Arrange the eigenvalues of A^TA in decreasing order. In Example 1, the eigenvalues are already listed in decreasing order: 360, 90, and 0. The corresponding unit eigenvectors, v_1, v_2, and v_3, are the right singular vectors of A. Using Example 1, construct

$$V = [v_1 \ \ v_2 \ \ v_3] = \begin{bmatrix} 1/3 & -2/3 & 2/3 \\ 2/3 & -1/3 & -2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix}$$

- The square roots of the eigenvalues are the singular values:

$$\sigma_1 = 6\sqrt{10}, \quad \sigma_2 = 3\sqrt{10}, \quad \sigma_3 = 0$$

THE SINGULAR VALUE DECOMPOSITION

- The nonzero singular values are the diagonal entries of D. The matrix Σ is the same size as A, with D in its upper left corner and with 0's elsewhere.

$$D = \begin{bmatrix} 6\sqrt{10} & 0 \\ 0 & 3\sqrt{10} \end{bmatrix}, \qquad \Sigma = [D \;\; 0] = \begin{bmatrix} 6\sqrt{10} & 0 & 0 \\ 0 & 3\sqrt{10} & 0 \end{bmatrix}$$

- ***Step 3*. *Construct U*.** When A has rank r, the first r columns of U are the normalized vectors obtained from $Av_1, \ldots, A\mathrm{v}_r$. In this example, A has two nonzero singular values, so rank A = 2. Recall from equation (2) and the paragraph before Example 2 that $\|Av_1\| = \sigma_1$ and $\|Av_2\| = \sigma_2$.

THE SINGULAR VALUE DECOMPOSITION

- Thus

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{6\sqrt{10}} \begin{bmatrix} 18 \\ 6 \end{bmatrix} = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

$$u_2 = \frac{1}{\sigma_2} A v_2 = \frac{1}{3\sqrt{10}} \begin{bmatrix} 3 \\ -9 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}$$

- Note that $\{u_1, u_2\}$ is already a basis for $\mathbb{R}^2$. Thus no additional vectors are needed for U, and $U = [u_1 \;\; u_2]$. The singular value decomposition of A is

$$A = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \begin{bmatrix} 6/\sqrt{10} & 0 & 0 \\ 0 & 3/\sqrt{10} & 0 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ -2/3 & -1/3 & 2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

THE SINGULAR VALUE DECOMPOSITION

- **Theorem: The Invertible Matrix Theorem (concluded)**
- Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.

u. $(\text{Col } A)^{\perp} = \{0\}$.

v. $(\text{Nul} A)^{\perp} = \mathbb{R}^2$

w. $\text{Row } A = \mathbb{R}^n$

x. A has n nonzero singular values.

THE SINGULAR VALUE DECOMPOSITION

- **Example 7** (Reduced SVD and the Pseudoinverse of A) When Σ contains rows or columns of zeros, a more compact decomposition of A is possible. Using the notation established above, let r = rank A, and partition U and V into submatrices whose first blocks contain r columns:

$$U = [Ur \quad Um\,\_\,_r], \quad \text{where } U_r = [\mathrm{u}_1 \ldots \mathrm{u}_r]$$

$$V = [Vr \quad Un\,\_\,_r], \quad \text{where } V_r = [\mathrm{v}_1 \ldots \mathrm{v}_r]$$

- Then U_r is $m\times r$ and V_r is $n\times r$. Then partitioned matrix multiplication shows that

$$A = [U_r \;\; Um\,\_\,_r]\begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} V_r^T \\ V_{n-r}^T \end{bmatrix} = UrDV_r^T \qquad (9)$$

THE SINGULAR VALUE DECOMPOSITION

- This factorization of A is called a **reduced singular value decomposition** of A. Since the diagonal entries in D are nonzero, D is invertible. The following matrix is called the pseudoinverse (also, the **Moore-Penrose inverse**) of A:

$$A^{+} = V_r D^{-1} U_r^T \qquad (10)$$